SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct:
I K. Ragavi (192524364) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified
I understand that any violation of academic integrity rules will result in for this assessment disciplinary action.

Signature of the student: K. Ragavi

Assessment Tasks

Application Based Questions

Question 1

A company experiences unpredictable traffic spikes during seasonal sales.

- Explain how cloud auto-scaling and load balancing can handle this demand efficiently.
- How does this approach reduce downtime and improve user experience?
- What role do monitoring tools play in this setup?
- Relate your answer to a real-world e-commerce scenario.

Answer

1. Auto-Scaling and Load Balancing for Traffic Spikes

- **Auto-scaling:** The cloud automatically adds virtual machines or containers when demand rises and removes them when it falls. Scaling policies are based on metrics such as CPU use, requests per second or response time. Scheduled scaling can pre-warm capacity before a known sale, and the company pays only for the capacity actually used.
- **Load balancing:** A load balancer is the single entry point for users and spreads requests across all healthy servers using algorithms such as round robin or least connections. It

checks server health and stops sending traffic to failed instances, and newly launched instances are registered with it automatically.

- **Together:** Auto-scaling supplies the capacity and the load balancer distributes work across it, so no single server is overloaded.

2. Reducing Downtime and Improving User Experience

- Servers are spread over multiple Availability Zones, so a zone failure does not stop the site.
- Failed servers are removed and replaced automatically, avoiding manual intervention.
- Capacity matches demand, so pages load quickly and checkout does not time out, even during peak hours.
- Stateless design with shared session storage (e.g., Redis) lets any server handle any user.
- Caching and a CDN offload static content and reduce load on the application servers.

3. Role of Monitoring Tools

Monitoring tools (Amazon CloudWatch, Azure Monitor, Prometheus/Grafana) collect metrics and logs such as CPU, memory, latency, error rate and request count. These metrics trigger the auto-scaling policies, drive dashboards for the operations team, and raise alerts through email or SMS when thresholds are crossed. They also help find bottlenecks, such as a slow database, and the historical data supports capacity planning for the next sale.

4. Real-World E-Commerce Example

During a Big Billion Days or Black Friday sale, an online store may see traffic rise ten times within minutes of the sale announcement. The store runs its web tier in an auto-scaling group behind an Application Load Balancer across three zones. When CPU passes 60%, new instances launch within minutes, while the CDN serves product images. If one instance crashes, the load balancer reroutes customers to healthy servers and the group launches a replacement. After the sale, extra instances are terminated, so the company is not paying for idle servers. Customers experience fast pages and a working checkout, and the business avoids lost sales.

Question 2

An organization wants to store large volumes of backup data securely in the cloud.

- Discuss how cloud storage solutions ensure durability and data redundancy.
- How can encryption and access control protect sensitive backup data?
- Explain the cost advantages compared to traditional storage systems.
- Provide an example of a disaster recovery use case.

Answer

1. Durability and Data Redundancy

- **Replication:** Cloud object storage (Amazon S3, Azure Blob, Google Cloud Storage) automatically stores multiple copies of each object across several devices and Availability Zones.
- **Durability:** Providers advertise very high durability, for example 99.999999999% (eleven nines) for S3, meaning loss of data is extremely unlikely. They use checksums to detect and repair corruption automatically.
- **Geo-redundancy:** Cross-region replication keeps a copy in another geographic region in case an entire region fails.
- **Versioning and immutability:** Versioning preserves earlier copies; object lock (WORM) stops deletion or overwriting, which also protects against ransomware.

2. Encryption and Access Control

- **Encryption at rest:** Backups are encrypted with AES-256, using provider-managed keys or customer-managed keys from a key management service (KMS/HSM).
- **Encryption in transit:** TLS protects data while it is uploaded or restored.
- **Access control:** IAM policies follow least privilege, so only backup roles can write and only a few administrators can restore or delete. Multi-factor authentication, MFA-delete, bucket policies, and private network endpoints reduce exposure.
- **Audit logging:** Every access is logged (e.g., CloudTrail) so that suspicious activity is detected.

3. Cost Advantages over Traditional Storage

Aspect	Traditional storage	Cloud storage
Upfront cost	High capital expense for hardware, data center space, power and cooling	No upfront cost; pay-as-you-go operating expense
Scalability	Must buy capacity in advance; slow to expand	Virtually unlimited and instantly scalable
Maintenance	Staff needed for hardware replacement and tape handling	Managed by the provider
Redundancy	Second site must be built and run by the company	Built in, at low additional cost
Tiering	Limited	Cheap archive tiers (e.g., Glacier) for rarely used backups

4. Disaster Recovery Use Case

A hospital backs up its patient database nightly to cloud storage with cross-region replication and object lock. If ransomware encrypts the on-premises servers or a fire destroys the data center, the IT team provisions virtual servers in the cloud from infrastructure templates, restores the latest clean backup and redirects users. Operations resume within hours, with data loss limited to the time since the last backup. The immutable copy cannot be altered by the attacker.

Question 3

A mobile app company plans to deploy its application globally.

- Explain how content delivery networks (CDNs) improve performance and reduce latency.
- How do edge servers enhance user experience across regions?
- Discuss how cloud providers support global availability.
- Relate this to a real-time streaming or gaming application.

Answer

1. How CDNs Improve Performance and Reduce Latency

A Content Delivery Network is a geographically distributed network of servers. Instead of every user fetching content from one distant origin server, the CDN stores cached copies of static content (images, videos, scripts, app updates) at locations close to users. The user is routed to the nearest location by DNS or anycast.

- Shorter distance means lower round-trip time and faster downloads.
- Cache hits are served without contacting the origin, which reduces origin load and bandwidth cost.
- Optimizations such as compression, HTTP/2 and persistent connections speed up delivery.
- Built-in DDoS protection and traffic absorption improve availability.

2. Role of Edge Servers

Edge servers sit in points of presence (PoPs) near end users. A user in Mumbai gets content from an Indian edge instead of a US server, giving quicker app start, smoother video and faster API responses. Edge servers can also terminate TLS connections early, run edge functions (e.g., Lambda@Edge, Cloudflare Workers) for authentication or personalization, and route requests to the best region. This gives consistent experience across regions regardless of location.

3. Global Availability Support by Cloud Providers

- Many regions and Availability Zones worldwide, so apps can be deployed near users and meet data-residency needs.
- Global load balancers and DNS services (Route 53, Azure Front Door) with health checks and latency-based routing.
- Multi-region database replication and global storage replication.
- A private global backbone network that is faster and more reliable than the public internet.
- Automatic failover to another region if one fails.

4. Streaming or Gaming Example

In a live video streaming app, the video is split into small segments cached at edge servers; viewers in each region receive them from a nearby PoP, which minimizes buffering, and adaptive bitrate adjusts quality to network speed. In an online multiplayer game, game assets and patches are delivered through the CDN, while matchmaking sends players to the nearest regional game server to keep ping low (for example under 50 ms). If a region fails, players are redirected to another region, so the game remains available worldwide.

Question 4

A financial institution requires strict compliance and data security in cloud usage.

- Explain how cloud providers support regulatory compliance and auditing.
- What role do logging, monitoring, and encryption play in security?
- How can the company ensure data sovereignty requirements are met?
- Give an example involving sensitive financial transactions.

Answer

1. Regulatory Compliance and Auditing Support

- **Certifications:** Major providers are audited against standards such as PCI DSS, ISO 27001, SOC 1/2/3, GDPR and regional banking rules, and publish compliance reports (AWS Artifact, Azure Compliance Manager).
- **Shared responsibility model:** The provider secures the infrastructure, while the institution secures its own data, identities, configuration and applications.
- **Compliance tools:** Services such as AWS Config, Azure Policy and Security Hub continuously check resources against compliance rules and flag drift.
- **Auditing:** Immutable activity logs (CloudTrail, Azure Activity Log) record every API call, supporting auditors and investigations.

2. Role of Logging, Monitoring and Encryption

- **Logging:** Provides a tamper-resistant record of who accessed what and when, which is essential for audit trails and forensics.
- **Monitoring:** Real-time monitoring and SIEM tools detect anomalies (unusual logins, large transfers) and trigger alerts or automated responses, supporting fraud detection and incident response.
- **Encryption:** AES-256 at rest and TLS in transit make stolen data unreadable. Keys are held in HSM-backed key management services with rotation and separation of duties; customer-managed keys keep control with the bank.

3. Ensuring Data Sovereignty

- Select cloud regions inside the required country or jurisdiction and use policies that block creation of resources in other regions.
- Keep backups and replicas within approved regions only.
- Use customer-managed or external key stores (BYOK/HYOK) held in the country.
- Check contracts and provider commitments on data location and government access; use sovereign cloud offerings where needed.
- Apply data classification and monitor data flows to prevent unauthorized cross-border transfer.

4. Example: Sensitive Financial Transactions

A bank runs an online payments service in a cloud region located in its home country. Card data is tokenized and encrypted with keys from an HSM-backed KMS. Each transaction travels over TLS, the application sits in a private network behind a WAF, and access to the database is

restricted by role and MFA. Every access and change is logged to an immutable store and monitored by a SIEM, which alerts on suspicious transfers. Replicas stay within the national region to meet sovereignty rules, and regulators can review the audit reports at any time.

Question 5

A development team wants to adopt containerization for application deployment.

- Explain how containers differ from virtual machines in terms of performance and resource usage.
- How does orchestration (like Kubernetes) simplify deployment and scaling?
- Discuss portability benefits across different cloud environments.
- Provide an example of a microservices-based application.

Answer

1. Containers vs Virtual Machines

Feature	Virtual Machine	Container
Isolation	Hardware-level virtualization via a hypervisor; each VM has its own guest OS	OS-level virtualization; containers share the host kernel
Size	Gigabytes	Megabytes
Startup time	Minutes	Seconds or milliseconds
Resource usage	High overhead (a full OS per VM)	Lightweight; many more per host
Performance	Slight overhead from the hypervisor	Near-native performance
Security isolation	Stronger isolation	Weaker, since the kernel is shared; needs hardening

Containers therefore give higher density and faster scaling, while VMs suit workloads needing strong isolation or different operating systems.

2. How Kubernetes Simplifies Deployment and Scaling

- **Declarative deployment:** The desired state is described in YAML, and Kubernetes makes the cluster match it.
- **Scheduling:** Places containers on suitable nodes based on resources.
- **Auto-scaling:** The Horizontal Pod Autoscaler adds or removes pods with load.
- **Self-healing:** Restarts failed containers and replaces unhealthy nodes.
- **Rolling updates and rollbacks:** Releases new versions with zero downtime.
- **Service discovery and load balancing:** Gives pods stable names and distributes traffic.
- **Configuration and secret management:** Separates settings from images.

3. Portability Across Cloud Environments

A container image packages the application with its libraries and dependencies, so it runs the same on a developer laptop, an on-premises server or any cloud (AWS, Azure, Google Cloud). This removes the "works on my machine" problem, eases migration and hybrid or multi-cloud adoption, reduces vendor lock-in, and allows consistent CI/CD. Kubernetes is available as a managed service on all major clouds, so the same manifests can be deployed almost unchanged.

4. Example of a Microservices-Based Application

An online food-delivery application is split into independent services: user service, restaurant/menu service, order service, payment service, delivery-tracking service and notification service. Each runs in its own container with its own database and communicates through REST APIs or a message queue behind an API gateway. During lunch hours, only the order and payment services are scaled up. A fault or update in the notification service does not affect the others, and each team can deploy its service independently using Kubernetes.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity